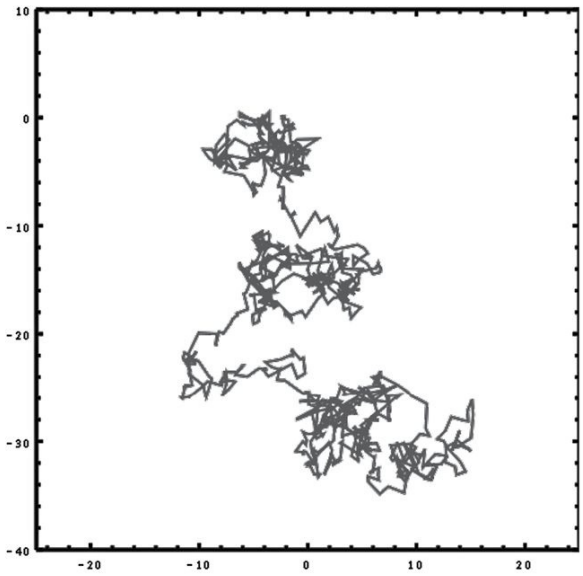


Figure 1.4 Typical Brownian motion.

The subtle point is the following. There are an enormous number of molecules of air. On average, as many hit the granule from the left as those that hit it from the right. If the air's molecules were infinitely small and infinitely numerous, the effect of the collisions from right and from left would balance and thus cancel out at each instant, and the granule would not move. But the finite size of the molecules – the fact that these are present in *finite* rather than infinite number – causes there to be *fluctuations* (this is the key word): that is to say, the collisions never balance out *exactly*; they only balance out *on average*. Imagine for a moment the molecules were very few in number and large in size: the granule would clearly receive a blow only occasionally; now one on the right, then one on the left … Between one collision and the other it would move here and there to a significant degree, like a football kicked by boys running around a playing field. The smaller the molecules, on the other hand, the shorter the interval between collisions and the better the hits from different directions balance and cancel each other out. And the less the granule moves.

It is indeed possible, with a little mathematics, to work back from the amount of movement of the granule, which can be observed, to the dimensions of the molecules. Einstein does this, as I mentioned above, at the age of twenty-five. From observations of granules drifting in fluids, from the measurement of how much these 'drift' – that is, move away from a position – he calculates the dimensions of Democritus's atoms, the elementary grains of which matter is made. He provides, after 2,300 years, the proof of the accuracy of Democritus's insight: matter is granular.